



Harnessing AI to Increase Workplace Safety

WorkplaceNL

WorkplaceNL Research Initiatives Program

Canadian
Home Builders'
Association





Contents

Contact Information	4
<i>Project Investigator</i>	<i>4</i>
Proposal Introduction	5
Scope and Research Objectives	6
Literature Review	7
Relevance and Significance of the Research	9
Research Plan and Methodology	10
Project Plan	12
Project Budget	13
Knowledge Transfer	14
Non-Scientific Summary	15
Potential Barriers	15
Qualifications, Relevant Experience, and Roles of the Research Team	16

Contact Information

Applicant Name

CHBA-NL

All Investigators

Training Works Inc

Primary Contact Person

Alexis Foster

Organization Name

Canadian Homebuilders Association – NL

Email

ceo@chbanl.ca

Mailing Address

48 Welland Street, St. John's, NL A1E6C1

Project Title

Harnessing AI to Increase Workplace Safety

Project Investigator



TRAINING WORKS

KNOWLEDGE BY DESIGN

Proposal Introduction

Please include a brief introduction that includes the following:

- A brief description of the study including the research question(s), issues or problem and how these will be addressed in the study.
- The relevance and benefits to Newfoundland and Labrador Workplaces and fit with WorkplaceNL's priority topic(s) or mandate.
- Steps to transfer research findings to end users.
- Funding amount requested.
- Duration of the study.

Workplace safety has made significant improvements through the adoption of engineered improvements and legislation. Despite these measures and the efforts of workplace safety organizations and industry associations, incidents and accidents still occur. Through a collaborative effort, CHBA-NL and Training Works seek to better understand the behavioural causes of incidents and accidents and through machine learning and a sound understanding of the operational environment, determine where an incident or accident is likely to occur. This work aims to enhance overall safety culture in the residential construction sector, decreasing the number of serious accidents.

To ensure lasting impact on safety culture within the residential construction sector, our team will use a strategic approach to disseminate knowledge and findings from this study. Various methods will be used for dissemination based on the audience, goal, and information to be provided. The overall goal of the strategy is to ensure actionable knowledge is transmitted to industry in plain language. To this end, CHBA-NL and Training Works will use their existing networks and relationships within the residential construction sector at each stage of the knowledge dissemination process.

To complete this impactful work, CHBA-NL is seeking \$148,500 in funding. The study is designed to take place over 12 months during which we intend to use safety datasets in a new manner, adding to the existing body of knowledge on behaviour-based safety.





Scope and Research Objectives

What is the research question? Clearly specify the research objectives of the proposed research and provide a rationale for each of the research objectives. Explain the problem to be solved and how the proposed research will address it.

This project intends to determine the key behaviours prior to safety incidents and accidents within the residential construction sector by leveraging data and machine learning. We hypothesize there are key behaviours leading to safety incidents and accidents that with proper identification could enhance safety in the sector. Through uncovering these behaviours and/or patterns of behaviours and logically categorizing into a digital competency model, an evaluation can be completed to determine whether there are existing skills gaps in these areas, thus validating the results of the machine learning algorithm.

Therefore, the research question(s) our project aims to answer are:

- Can workplace safety incidents within the residential construction sector be linked to specific behaviours and/or patterns of behaviours?
- Can an algorithm be developed to analyze safety reports to identify the behaviours leading to incidents/accidents?

Literature Review

Provide a critical review of the existing research, as well as knowledge and literature for the proposed area of research. Include an analysis of the relative strengths, weaknesses, limitations and gaps in knowledge relative to the proposed area of research.

Workplace safety incidents are a significant global issue, and in particular, the construction industry is over-represented in workplace injury and death statistics (Biggs & Biggs, 2013). The International Labour Organization (2015) reports at least 60,000 fatal accidents occur each year at construction sites worldwide, equivalent to one death every 10 minutes. In the research literature, a 2013 special issue in the journal of Construction Management and Economics devoted an entire issue to focus on addressing health and safety in the construction sector, signaling its importance. Further, interest in safety behaviours within the construction industry has attracted meta-analytical investigations exploring the links between safety critical tasks, competencies with which they are performed, and their overall impact on safety cultures (Christian et al., 2009). Remediation strategies aimed at modifying workplace behaviour(s) leading to accidents is a high priority for organizations since research has identified that a positive safety culture (i.e., one that promotes and rewards safe behaviours) is correlated with positive improvements to traditional lag indicators such as injury and time off work (Dodsworth et al., 2007; Silva et al., 2004). The behaviours workers perform in their daily jobs can have a direct and immediate effect on their health and safety (Guo et al., 2018), and researchers have employed various methods for identifying unsafe behaviour and studying how to reduce such behaviour(s) on job sites (Choudhry & Fang, 2008; Shin et al., 2015; Guo et al., 2016).

DeJoy (2005) outlined two approaches to managing workplace safety, namely organizational culture and behaviour-based approaches. Behaviour based safety (BBS) approaches are 'bottom-up' approaches focusing on the identification and modification of critical safety behaviours and emphasizes how such behaviours are linked to workplace injuries and losses. BBS management is essentially an extension of applied behaviour analysis or behaviour modification (Elder et al., 1994), where key or critical behaviours are objectively identified and targeted for change. BBS has its origins from behaviourism (also known as behaviour modification, operant psychology, and applied behavioural analysis) (see DeJoy, 2005), and there has been abundant evidence that BBS initiatives are effective in reducing unsafe behaviours (Duff et al., 1994; Krause et al., 1999; Choudhry, 2014).

Despite a growing appreciation that workers are an untapped source of good ideas for innovative and effective occupational health and safety solutions, management structures remain hierarchical and focus primarily on top-down behavioural control (Sherratt et al., 2013). We see this single-viewed approach to addressing workplace safety as a limitation, and our research considers the value of bottom-up BBS approaches. Consulting front line workers for insights into what behaviours and human factors are most likely to lead to injuries, incidents and near misses has been under investigated in the construction industry.

The current research focuses on BBS approaches to improving workplace safety within the construction industry. Our project leverages the advantages of using machine learning to sort complex incident reports and analyze their relations to workplace behaviours. Such technological approaches to analyzing unstructured data, also regarded as data mining (see Feldman, 2007), offers promising directions to understand why and how incidents happen in the workplace (Nakata, 2020).

Currently in the construction industry, traditional performance indicators such as lost time injuries (LTIs) and related lag indicators are the primary measure of occupational health and safety performance. However, these measurements can only be collected after an incident or accident has occurred (Biggs & Biggs, 2013). Our plan is to take a proactive approach using datasets and machine learning to identify the behaviours most likely to lead to workplace incidents and accidents. These behaviours will be correlated to Training Works' digital competency model which can be used to assess skills gaps within the residential construction sector allowing for remediation measures to be put in place, thus perpetuating safety culture. A strong safety culture is proactive about safety rather than reactive and helps ensure safer, more productive job sites by supporting a strong safety climate (Dodge Data & Analytics, 2016).



Relevance and Significance of the Research

Explain how the proposed research would be different from previous and on-going studies identified in the literature review. Include the problems that the proposed research will address including any gaps in knowledge, policy, programs or practices.

Describe the project's purpose and relevance as it relates to WorkplaceNL's mandate including the anticipated benefits to Newfoundland and Labrador's workplaces and stakeholders.

To prevent workplace injury and illness we must first identify if and what behaviours are causing them. Through this research our team will leverage big data and AI to investigate the behaviours and patterns of lagging indicators to understand and identify any current skills gap. Identifying key behaviours and potential skills gaps will give great insight into the current safety culture and the operational reality of working in the residential construction sector. A strong safety culture is rooted in ownership and accountability, individuals and organizations alike must own their safety and be accountable for their actions and behaviours. Understanding this current reality sets a strong foundation for increasing the safety culture of workplaces in Newfoundland and Labrador and preventing serious injuries. Research has identified that positive safety culture is correlated with positive improvements to traditional lag indicators such as injury and time off work (Dodsworth et al., 2007, Silva et al., 2004).

According to the Construction Owners Association of Alberta (COAA) in 2008 approximately 90% of the incidents analyzed involved behaviour as a factor. (COAA, 2008). Previous research in this area has suggested that "accidents" are a result of eight factors: lack of proper training; deficient enforcement of safety rules; lack of safety equipment; unsafe work methods; unsafe site conditions; failure to use proper safety equipment; poor attitudes held towards safety; and isolated unavoidable causes (Toole, 2002). This research is somewhat dated and many of these factors are not related to human behaviours, which the COAA have demonstrated in their review of incidents are a primary contributor factor (COAA, 2008). This research project's goal of identifying and mapping those behaviours linked to workplace incidents and accidents with the goal of enhancing safety culture and reducing and preventing injury and illness is different from the previous studies found in the residential construction sector literature because of its focus on behaviour.

Research Plan and Methodology

Provide a clear explanation of the theoretical/conceptual framework guiding the research with the minimum requirements:

- Rationale for selection of the theoretical framework.
- Research methodology: what methods will be used and why are they appropriate? Explain how the research will be conducted.
- Sample information: description, estimated size, who the participants will be and how are they selected, when and where the study will be done. Clearly detail who will be recruiting participants and obtaining consent, and the procedures for doing this.
- Include your approach to data collection and analysis. Identify data sources and software. What material or instruments will be used to collect data? Indicate if quantitative or qualitative methods of analysis are planned.
- Indicate if there are any ethical considerations.
- Outline the timelines and deliverables in a table format. Include key activities and the person responsible, with start date and end dates.

Based on the literature review, it is apparent data collected from safety incident and accidents is leveraged primarily for roll-up reporting and top-down management. Our research is grounded in behaviour-based safety (BBS) interventions (Geller, 1999; 2001) which concentrate on workers' behaviour as the source of most work-related accidents, it is our hypothesis that skills gaps in specific behaviours cause safety incidents and accidents. These behaviours can be influenced by such moderating factors as education, training, coaching and mentoring, but first, they must be identified in order to properly implement remediation strategies. Moreover, lack of identification and remediation of these behavioural factors in the moment increase the risk factor for safety incidents/accidents.

Through application of Machine Learning, we intend to apply a proprietary algorithm to identify the patterns in behaviour leading to incidents and accidents. It is our theory this algorithm can review safety data on a continuous basis and highlight patterns of behaviour which appear to have a causal relationship with safety incidents and accidents. This data could serve as the new standard for safety culture by identifying

leading indicators which when provided to management allow for safety remediation before an accident occurs, thus decreasing the number of jobsite incidents and accidents within the residential construction sector.

Our research plan includes a mixed methods approach (Bryman et al., 2011), leveraging the breadth of quantitative secondary data (i.e., archived incident reports) and the depth of qualitative semi-structured interviews.

Key informant interviews will be administered using a semi-structured approach (Bryman et al., 2011). In his description of the benefits of interviews, McCracken (1988: 9) maintained "The method can take us into the mental world of the individual, to glimpse the categories and logic by which he or she sees the world . . . can also take us into the lifeworld of the individual". For these reasons we believe such an interpretivist approach, is best suited in surfacing the key behaviours linked to workplace incidents in the sector. Using a bottom-up approach, participants will share their experiences of operational realities working in the sector helping to inform the refinement of the ML process.

Participants will be selected from registered members of CHBA-NL. We will seek a target sample size of 40-50 participants, collected across several companies. Our participants will be selected through management network and a 'snowballing' process (Bryman et al., 2011). The process and requirements will be clearly discussed with each participant prior to interview process allowing them the opportunity to decide whether they would like to participate.

Personalized data collected throughout this project is for the purposes of evaluation and continuous learning. No personalized data will be shared with WorkplaceNL and only data in aggregate form will be provided as part of the final reporting to WorkplaceNL. There will be no research activities conducted by WorkplaceNL using personalized data. Therefore, according to TCPS 2 Chapter 2: Scope and Approach (2018)

Quality assurance and quality improvement studies, program evaluation activities, and performance reviews, or testing within normal educational requirements when used exclusively for assessment, management or improvement purposes, do not constitute research for the purposes of this Policy, and do not fall within the scope of REB review. (Article 2.5, para. 1)

Once all interviews are completed, the data will be analyzed using a two-step process, namely open coding software analysis and then categorization using theoretical saturation. This process of data analysis will rely on the basic premises of content and thematic analysis (Glaser and Strauss, 1967), interview analysis (Kvale, 1996) and the use

of data reduction methods (Huberman and Miles, 1994). Qualitative data analysis is a subjective process that originates between the researcher(s), the data and the interviewee. Therefore, we will make all efforts to reduce biases in our coding process.

In tandem with conducting key informant interviews, we will employ a quantitative research approach to assess various sources of company, WorkplaceNL, and the NL government's secondary archival data. These reports may help isolate whether certain patterns are more common than others. Note: The exact datasets required will be determined in the early stages of the project. Risks and mitigation strategies associated with this data collection are outlined in the Potential Barriers section of this application.

Our model will then compile the accident report data, identify the cause, and categorize the accidents as either threat or error. Once this is complete the accidents identified as errors (behaviour based) will be subject to further analysis for our data synthesis and those that fall into the threat category will be compiled as statistics.

Note: If the appropriate data sets are not in numerical representations, the team will harvest and index content, keywords, and metadata from the source safety datasets. The data will then be methodologically categorized in consultation with Subject Matter Experts.

The patterns in behaviours will be documented and categorized into Training Works' digital BBS competency model which will provide a mechanism for determining skills gaps in key areas. Over time, and outside scope of this funding request, the model can be used to push out learning content to remediate skills gaps.

Project Plan

Activity	Milestone/Deliverable	Date	Responsible Party
Contract Award	Signed Contract	6/30/21	CHBA-NL/WorkplaceNL
Project Kick Off Meeting	Project Charter	7/7/21	CHBA-NL/Training Works
Planning the Study - Identifying needs for study - Refine research questions - Define study protocol	Study Protocol	7/7/21 - 7/21/21	Training Works
Collect safety incident and accident data - Collect CHBA-NL data - Collect WorkplaceNL data - Collect OHS data from NL government	Compilation of available pertinent data	7/21/21 - 9/8/21	CHBA-NL/Training Works
Conduct Primary Research - Conduct interviews with participating companies - Develop and deploy survey to participating companies	Contextualization of safety incident and accident data	9/8/21 - 2/16/22	CHBA-NL/Training Works
Assimilate Data into Model - Harvest/index content, keywords, and metadata from source datasets - Logically categorize based on industry experience and primary research - Establish baseline of key - Train ML module(s)	Training of ML Module(s)	9/8/21 - 11/10/21	Training Works
Information Synthesis and Build out of System - Document and categorize patterns of behaviours into digital competency model	Development of data driven digital competency model	11/10/21 - 3/16/22	Training Works
Evaluate Skills Gaps	Utilization of competency model within identified participant companies	3/16/22 - 4/15/22	Training Works
Mapping and Documenting Results	Final Report	4/15/22 - 5/31/22	Training Works
Knowledge Dissemination	Completion of identified knowledge dissemination activities	5/31/22 - 6/30/22	CHBA-NL/Training Works

Project Budget

Please provide a description and justification of each budget item and include a table outlining budget categories.

The following table outlines the budget breakdown for this project. The budget is comprised of a combination of existing team consultants slated to complete the research along with salaried consultants. Note, the principal investigators salary is not included in this budget per WorkplaceNL Research Initiatives guidelines. The machine learning line item refers to the costs associated with machine learning module training, information synthesis and buildout as represented in the workplan defined in the previous section. A comprehensive breakdown of remaining line items is included in the table below.

Salaries				
	Positions	Daily Cost	Days	Total
	Project Manager	\$300.00	25	\$7,500.00
	Learning designer	\$300.00	60	\$18,000.00
	Researcher	\$250.00	100	\$25,000.00
	media developer	\$215.00	40	\$8,600.00
Salaried Contractors				
	Positions	Hourly Rate	Hours	Total
	Senior Researcher	\$225.00	20	\$4,500.00
	Behaviour Based Safety Specialist	\$250.00	20	\$5,000.00
Machine Learning				
Overhead				
	Project Overhead/ Management for CHBA			\$15,000.00
	Professional Fees			\$10,000.00
	Photocopying/Office Expenses			\$500.00
Equipment				
	Tablets (\$400 each)			\$4,000.00
Knowledge Dissemination				
	Presentations			\$500.00
	Case Study			\$500.00
	Short Video Series			\$1,000.00
	Infographic(s)			\$250.00
	Online Campaign			\$250.00
Total				\$148,500.00

Knowledge Transfer

Identify the products, methods and key user groups or potential audiences that would be interested in your research findings. Include whether you plan to publish your findings and in which publications, as well as how you will enable the usability and transferability of your research findings. Clear language format is required for final reports, training materials and resources used for non-academic audiences. WorkplaceNL also requires permission to post final reports, PowerPoint presentation with notes, and a 2-3-page executive summary that focusses on findings as opposed to methodology.

Creating an impactful Knowledge Transfer (KT) plan involves a strategic approach which includes the resources and expertise within the budget and a KT timeline. However, it also involves understanding the audience, the goal of the KT, the information to be shared and the methods used. Our goal is to impact industry, to go beyond the existing safety training and policies and provide insight into the behaviours driving incidents and accidents in the workplace.

The first level of knowledge transfer is for awareness, this audience is large and involves members of CHBA-NL and other provincial chapters, safety groups, and additional interested parties. This type of KT includes a brief overview of the project and findings where it is included in presentations, a brief description on Training Works' website, social media and mentioned in discussions with similar organizations.

The second level is for real understanding of the project and research findings with the audience benefiting from the results. At this level, our audience includes project stakeholders and organizations providing information and support. This audience is provided detailed information on methodology, research findings, conclusion and validation information through presentations and report format along with a short video series describing the findings.

To impact industry there must be inspired action. While gaining access to research findings, this audience can leverage the identified behaviours for new skills development within the industry to ultimately prevent and reduce workplace incidents and accidents. This final level of KT involves all identified channels and methods, an online strategy including distribution of the short video series through company and organization websites, social media, newsletters, infographics, along with virtual or in-person presentations, and an industry case study.

Training Works will avail of their existing network and relationships within the residential construction industry at each stage of KT and information sent through CHBA-NL's membership, website, and social media channels and amplified through Training Works' channels.

Non-Scientific Summary

Include a brief non-scientific summary describing the project issue, the project objectives, and the expected impact of your findings that WorkplaceNL could use for release to a wider audience (i.e., posted on WorkplaceNL.ca).

In 2019, the construction industry had the highest number of workplace fatalities in Canada (Fatalities in Canada – 2019, 2021). Given the disproportionate numbers in this industry, a new way of predicting and preventing incidents and accidents is required to ensure everyone goes home safely.

Our project aims to leverage safety data to train a machine learning algorithm on the key behaviours occurring immediately before an incident or accident. Over time, this algorithm will be trained to spot combinations of behaviours which indicate an incident or accident is on the horizon and to provide alerts to key safety personnel allowing for intervention. The underpinning of this concept is a data driven behaviour-based safety competency model which outlines and evaluates the key behavioural factors impacting safety.

The research completed in this project will assist companies in spotting and stopping unsafe behaviours in the moment they occur, preventing incidents and accidents from happening. Using leading indicators, it is our belief the residential construction industry can reduce the number of workplace fatalities and lost time injuries which are caused by behavioural factors.

Potential Barriers

Are there any challenges you foresee to completing this project in the proposed time period? If so, please describe them and your plan to mitigate them.

The project team has identified access to data as the highest-level risk associated with the project. As a mitigation strategy, the Training Works team met with Mr. Loyola Power, Director of Occupational Health and Safety at the OHS Division of Digital Government and Service NL to discuss the project. Mr. Power provided a letter of support (attached). The team also met with Ms. Cathy Whiffen, Director of Prevention Services at WorkplaceNL. To effectively mitigate this risk, the team is prepared to leverage qualitative data from companies combined with root cause analysis reports and WorkplaceNL/OHS data to develop a comprehensive risk profile for the industry.

The project team has also identified lack of buy-in as a high-level risk associated with project success. Ensuring bidirectional support for this project will be critical to success since the team will rely on individual companies to provide qualitative data for analysis and research to help build the industry risk profile. To mitigate this risk, a comprehensive value proposition targeting all levels of the organization will be developed. Through this effort, company owners, forepersons, and construction workers will understand the importance of creating a risk profile as part of a larger scale plan to decrease safety incident and accidents within the sector.

Qualifications, Relevant Experience, and Roles of the Research Team

Please outline research team information and attach CVs (please add rows if necessary).

Research Team Information	
Name	Tamara Vatcher
Title	Partner
Organization	Training Works
Department	Operations
Address	P.O. Box 14141, Stn A St. John's, NL A1B 4G8
Telephone	709.685.8209
Email	Tamara@training-works.ca
Name	Jennifer Reddy
Title	Project Specialist
Organization	Training Works
Department	Operations
Address	P.O. Box 14141, Stn A St. John's, NL A1B 4G8
Telephone	709.222.3848
Email	Jennifer@training-works.ca
Name	Rene Arsenault
Title	Research Specialist
Organization	Training Works
Department	Operations
Address	P.O. Box 14141, Stn A St. John's, NL A1B 4G8
Telephone	902.997.1993
Email	Rene@training-works.ca
Name	Branden On
Title	Multimedia Specialist
Organization	Training Works
Department	Operations
Address	P.O. Box 14141, Stn A St. John's, NL A1B 4G8
Telephone	709.727.4963
Email	Branden@training-works.ca

Research Team Information (continued)	
Name	Susan Murray
Title	Senior Research Consultant
Organization	Training Works
Department	Operations
Address	P.O. Box 14141, Stn A St. John's, NL A1B 4G8
Telephone	709.697.3502
Email	Smurray92@gmail.com
Financial Officer Information	
Name	Alexis Foster
Title	Executive Officer
Organization	CHBA-NL
Address	48 Welland Street St. John's, NL A1E6C1
Telephone	709.753.2000
Email	Ceo@chbanl.ca
Research Proposal Signatures	
Principal Applicant	
Name	Alexis Foster
Title	Executive Officer
Organization	CHBA-NL
Address	48 Welland Street St. John's, NL A1E6C1
Telephone	709.753.2000
Email	Ceo@chbanl.ca
Date	April 30, 2021
Signature	See attached WorkplaceNL application template for signature
Co-applicant (if applicable)	
Name	
Title	
Organization	
Address	
Telephone	
Email	
Date	
Signature	

**Canadian
Home Builders'
Association**

